

Logic in Computer Science, Prolog

BITS Pilani- K K Birla Goa Campus

Total: 16 Marks

Weightage: 8%

Course No: *CS F214*

Time : 45 Minutes

December 09, 2022

Note:

- Store your file using your ID number. For example if your ID no is 2014A7PS1234G then your file name should be 2014A7PS1234G.pl
 - Your file should contain four predicates (ackermann/3, sublist/2, triplets/3, powers/3).
 - You have to provide definitions of all the auxiliary predicates used and you are not allowed to use any libraries.
 - There should be no compilation error when you run your prolog code.
 - The test cases might have variables as well not only just ground queries. For example, one test case for triplets may be triplets(X,Y,Z).
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1. Write a prolog program with the following predicates

- (a) reverse(L1,L2) returns true iff the list L2 is the reverse of the list L1. (2 Marks)

For example, reverse([2,3,1],[1,3,2]) returns true and reverse([3,7,1,9],[1,7,9]) returns false.

- (b) ackermann(X,Y,Z) returns true iff $Z = a(X,Y)$ where a is the Ackermann's function from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$ is defined by the following equations. (2 Marks)

$$a(0,y) = y$$

$$a(x,0) = a(x-1,1)$$

$$a(x,y) = a(x-1, a(x,y-1))$$

For example, ackermann(1,2,4) returns true and ackermann(0,3,5) returns false.

- (c) sublist(L1,L2) returns true iff the the list L2 is the sublist of the list L1 (3 Marks)

For example, sublist([2,4,3,7,8],[4,3,7]) returns true and sublist([2,4,3,7,8],[4,7]) returns false.

- (d) triplets(X,Y,Z) returns true iff X, Y and Z are different integers between 0 and 9 both included and $\frac{10 \cdot X + Y}{10 \cdot Y + Z} = \frac{X}{Z}$ with infinite precision. (4 Marks)

For example, triplets(1,6,4) returns true and triplets(2,4,6) returns false.

- (e) powers(L1,N,L2) returns true iff the list L2 contains the smallest N integers that are a positive (non-zero) power of one of the elements of L1. (5 Marks)

For example powers([3,5,4],10,L) returns $L = [3,4,5,9,16,25,27,64,81,125]$ and powers([4,2],6,L) returns $L = [2,4,8,16,32,64]$

Hint: Solve by generating the first N powers of all the given numbers in L1, merge them into single list and the smallest N of them will be L2. It is not efficient and might not work for bigger L1 and N. This solution might get you partial marks only.